PRODUCT REQUIREMENTS DOCUMENT (PRD)

Multi-Mode Tournament Engine – Esports SaaS Platform (Updated)

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1. Executive Summary

The Multi-Mode Tournament Engine enables organizers to create and manage Free Fire tournaments across multiple competitive formats including Solo, Duo, Squad, Clash Squad, and Community Custom Rooms.

The system provides intelligent room assignment, automated scoring, customizable point configuration, secure multi-tenant isolation, moderator-level access control, and real-time leaderboard updates.

This module is designed for scalability, flexibility, and SaaS monetization readiness.

2. Objectives

- Support multiple Free Fire game modes.
- Enforce 48–50 player match limits.
- Allow organizers to customize scoring rules.
- Automate room assignment logic.
- Ensure strict tenant isolation.
- Support moderator-based kill updating.
- Enable real-time leaderboard updates.
- Scale to 10,000+ organizers.

3. Supported Game Modes

Solo Mode:
- 1 player per team.
- 48–50 players per match.
- Individual leaderboard.

Duo Mode:
- 2 players per team.
- Maximum 24–25 teams per match.

- Team-based leaderboard.

Squad Mode:
- 4 players per team.
- Maximum 12 squads per match (48 players).

Clash Squad Mode:
- 4v4 format.
- 2 squads per room.
- Round-based structure.

Community Custom Rooms:
- Admin defines player limit (up to 50).
- Hybrid team structures supported.

4. Adjustable Scoring System

The platform allows organizers to fully customize scoring rules per tournament.

Default Placement Points:
1st – 12
2nd – 9
3rd – 8
4th – 7
5th – 6
6th – 5
7th – 4
8th – 3
9th – 2
10th – 1

Default Kill Points:
1 point per kill

Customization Options:
- Organizer can modify placement points.
- Organizer can modify kill multiplier (e.g., 2 points per kill).
- Organizer can disable placement scoring (kill-only format).
- Organizer can define custom tie-breaker rules.
- Scoring configuration is locked once tournament starts.
- All scoring logic is stored per tournament in database.

Total Points Formula:
Total Points = Placement Points + (Kills × Kill Multiplier)

Tie Breaker (Default):
- Higher kills rank higher.
- If tied, earliest submission ranks higher.

5. Smart Room Assignment System

- Automatic grouping based on selected mode.
- Prevents room overflow beyond 50 players.
- Ensures no duplicate player assignment.
- Generates unique Room ID per match.
- Locks room once full.
- Logs all assignments for audit tracking.

6. Security & Multi-Tenant Isolation

- Separate database per organizer.
- Dynamic tenant connection middleware.
- No shared collections across tenants.
- Role-based access control (RBAC).
- Moderator restricted to assigned room only.
- Hashed access keys with expiration.
- Scoring settings isolated per tenant.

7. Scalability & Infrastructure

- Stateless backend architecture.
- MongoDB Atlas deployment.
- Redis for real-time scaling.
- Load balancer support.
- Rate limiting per tenant.
- Audit logging for critical actions.

8. Future Enhancements

- AI-based cheat detection.
- Public tournament marketplace.
- Entry fee integration.
- Advanced analytics dashboard.
- OBS/Twitch overlay API.
- Global player ranking system.